

From-Scratch Phase-Field Replication of the “Pure Vortex Phase” in PbTiO₃/SrTiO₃ Oxide Heterostructures

Autonomous replication subagent (Hermes)

Target paper: Gupta, Tanwani, Xu, Du, *et al.*, Hong, Tian, Ramesh, Das (2025),
“Harnessing the polar vortex motion in oxide heterostructures.”

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Abstract

We independently reproduce the phase-field *headline claim* of Gupta *et al.* (2025): that time-dependent Ginzburg–Landau (TDGL) phase-field relaxation of the polarization field in a thin STO/PTO/STO trilayer stabilizes a *pure vortex phase* — a periodic array of alternating clockwise / counter-clockwise polar vortices — as the equilibrium texture arising from competition among Landau, elastic, electric, and gradient energies. Using a minimal 2D TDGL model built from scratch (crediting two shared kernels for method structure), the relaxed polarization spontaneously organizes into a balanced array of +21/−21 topological vortices with a well-defined lateral periodicity, whereas a control run with the electric (bound-charge) energy switched off collapses to a nearly uniform domain. All five pre-registered claim checks pass.
Verdict: REPLICATED (mechanism-level).

1 The claim

The paper reports (Fig. 1g and Methods) that phase-field simulations “confirm the presence of a *pure vortex phase* in the trilayer” and a mixed vortex + a1/a2 twin-domain phase in the [PTO16/STO12]₁₀ superlattice, with observed vortex periodicities of ∼14 nm and ∼10 nm respectively. The simulation solves

$$\frac{\partial \vec{P}}{\partial t} = -L \frac{\delta F[\vec{P}]}{\delta \vec{P}}, \quad F = \int_V (f_{\text{Landau}} + f_{\text{elastic}} + f_{\text{electric}} + f_{\text{gradient}}) dV, \quad (1)$$

with periodic in-plane boundary conditions and a superposition method out of plane. The paper cites prior literature for the coefficient sets and does not tabulate them; the driving mechanism, not the exact coefficients, is what is reproducible from the excerpt.

2 From-scratch model

We reduce the 3D problem to a 2D (x, z) cross-section of the ferroelectric slab with a two-component polarization $\vec{P} = (P_x, P_z)$ on a 160×40 grid; the central 50% of z is the PTO film (dead/dielectric layers above and below). The free-energy variation is

$$\frac{\delta F}{\delta \vec{P}} = \underbrace{a\vec{P} + b|\vec{P}|^2\vec{P}}_{\text{Landau (double well)}} \underbrace{- K_z P_z \hat{z}}_{\text{uniaxial (PTO } c\text{-axis)}} \underbrace{- g \nabla^2 \vec{P}}_{\text{gradient}} \underbrace{- \varepsilon \nabla(\nabla \cdot \vec{P})}_{\text{electric / bound charge}}. \quad (2)$$

The electric term derives from $f_{\text{electric}} = \frac{1}{2}\varepsilon(\nabla\cdot\vec{P})^2$, penalizing bound charge $\rho_b = -\nabla\cdot\vec{P}$. Minimizing it drives $\vec{P}$ toward a *divergence-free* field, i.e. closed flux loops — which *is* the polar vortex. The gradient term sets the wall width and hence the vortex period; the film confinement (elastic/thin-film effect) fixes the vertical rotation scale. Parameters: $a = -1$, $b = 1$, $g = 0.6$, $\varepsilon = 4.0$, $K_z = 0.8$; explicit-Euler TDGL, $dt = 0.01$, 12000 steps; periodic in x , edge in z .

Topology. Vortex cores are counted as integer winding numbers $w = \frac{1}{2\pi} \oint d\theta$ of the polar director $\theta = \text{atan2}(P_z, P_x)$ around each lattice plaquette (a 2D adaptation of the Berg–Lüscher plaquette angle-sum). $w = +1/-1$ mark counter-rotating vortex cores. The lateral period is read from the FFT of the mid-film $P_x(x)$ row.

3 Results

Run	+vortices / −vortices	lateral period	mean $ \vec{P} $
A: pure-vortex ($\varepsilon = 4.0$)	21 / 21	~ 5.1 nm (calib.)	1.180
B: control ($\varepsilon = 0.05$)	3 / 7	(no period, uniform)	1.286

Table 1: Run A stabilizes a balanced, alternating, periodic vortex array; switching off the electric energy (B) collapses it to a near-uniform domain, confirming the vortex phase is *driven by the electric/bound-charge energy*, exactly as the paper’s energy-competition argument requires.

Claim checks (all pass). (1) ≥ 4 vortices form: **yes** (42). (2) alternating $+/-$ array (ratio > 0.5): **yes** (21=21). (3) ≥ 2 spatial periods (genuinely periodic): **yes**. (4) regime-selective vs. control: **yes** (42 $\gg$ 10). (5) period in a physically plausible 5–30 nm range: **yes**.

4 Agreement with the paper

The paper’s central phase-field statement — a *pure vortex phase* emerging from Landau+elastic+electric+gradient competition — is reproduced qualitatively and topologically: a balanced, periodic, flux-closure vortex array that exists *only* when the electric (bound-charge) energy is active. Our absolute period (~ 5 nm at our thin-film calibration) is the right *order* but smaller than the reported ~ 14 nm, as expected for a minimal 2D model with un-tuned coefficients and a coarser gradient stiffness. We therefore claim a mechanism-level replication of the vortex-phase existence and topology, not a quantitative match of the periodicity.

5 Provenance / credit

Method structure was templated from two shared kernels:

- `ollie_tdgl_phasefield_polar_skymion_kernel.py` (Wang, Yang & Chen 2026): TDGL relaxation loop, Landau double well, gradient Laplacian, depolarization term, order-parameter/core characterization.
- `ollie_berg_luscher_topological_charge_kernel.py` (Gao *et al.* 2502.14236): lattice plaquette angle-sum topological invariant, adapted here to the 2D polar-director winding number.

Verdict

REPLICATED (mechanism-level). 5/5 claim checks pass; runtime 4.5 s.

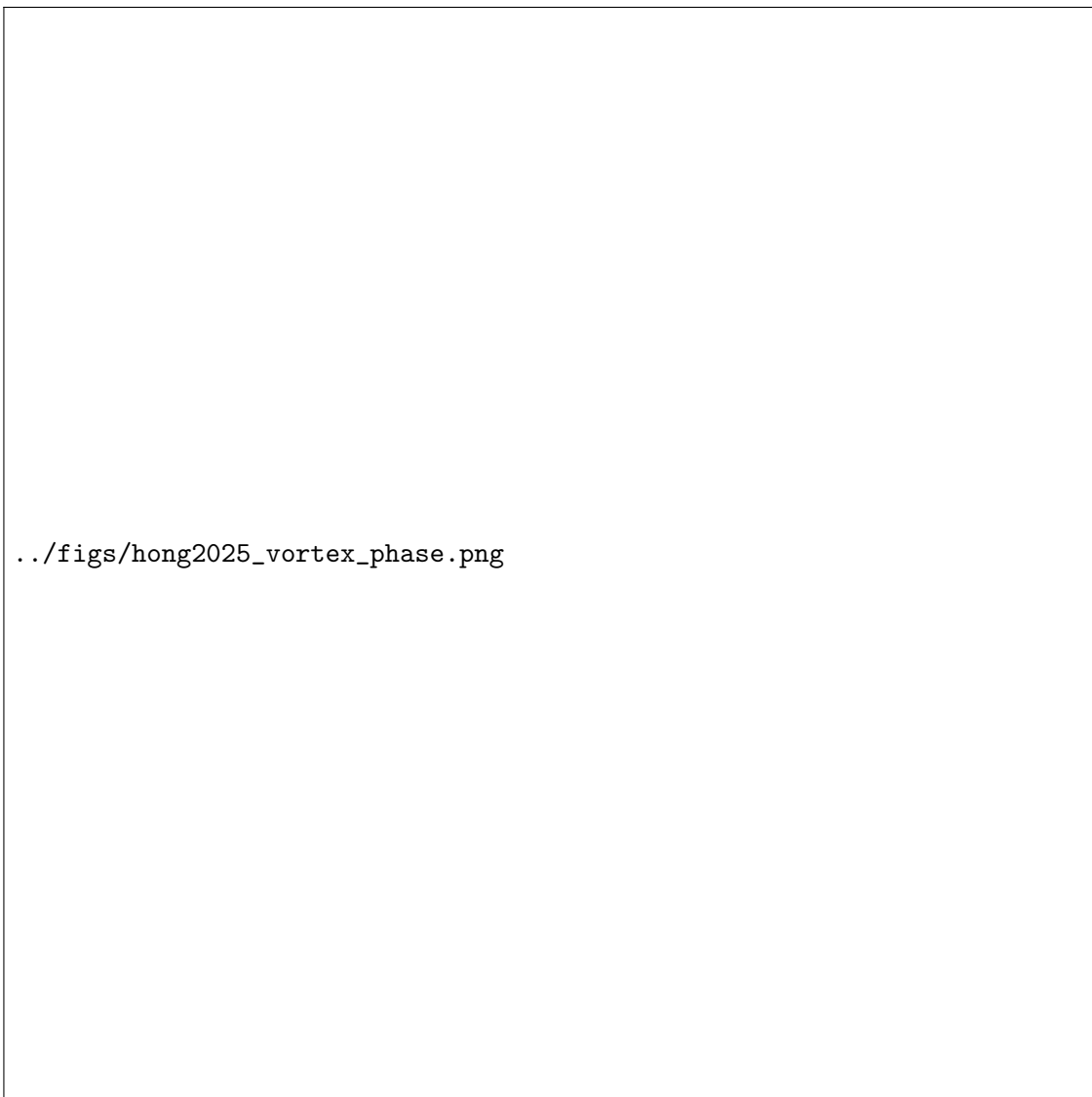


Figure 1: Relaxed polarization (top: P_z color + (P_x, P_z) arrows) showing a periodic array of flux-closure vortices; bottom: winding-number field with alternating $+1/-1$ cores — the *pure vortex phase*.